

DAY 66
14/09/2025

$$n = 5$$

7	—	7	—	0
7	—	7	—	7
7	—	7	—	0
7	—	7	—	7
7	—	7	—	7

The screenshot shows an IDE with two windows. The left window displays the source code for `Patterns2.java`. The right window shows the terminal output.

```

Patterns2.java | Patterns2.java 2, U x J > < ... 
ops > J Patterns2.java > Java Language Support > Patterns2 > PrintS

3
4 public class Patterns2 {
5
6     public static void printStarOdd(int n){
7
8         System.out.println("Printing Odd Pattern");
9
10        for(int i = 1 ; i<n ; i++){
11
12            for(int j = 1 ; j<n ; j++){
13
14                if((j&1) == 1){
15                    System.out.print(s:"* ");
16                }
17
18                else{
19                    System.out.print(s:" ");
20                }
21            }
22
23            System.out.println();
24
25            System.out.println();
26
27        }
28
29    }
30
31    Run [ Debug ] Run main [ Debug main
32    public static void main(String[] args) {
33
34        printStarOdd(n:7);
35    }
36 }

(base) kveeresh@KS-MacBook-Air-2 Computer-Science
• cd /Users/kveeresh/Desktop/Computer-Science ;
  /usr/bin/env /Users/kveeresh/Library/Java/JavaV
rtuaMachines/openjdk-24.0.1/Contents/Home/bin/ja
va -agentlib:jdwpt=transport=dt_socket,server=n,su
spend=y,address=localhost:53724 --enable-preview
-XX:+ShowCodeDetailsInExceptionMessages -cp /User
s/kveeresh/Library/Application_Support/Code/User
/workspaceStorage/2ecbf2b3509965b320b049db629e694
6/redhat_java/jdt_ws/Computer-Science_4e282deb/bi
n-Loops.Patterns2
Printing Odd Pattern
* * *
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* * *
* * *
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Skipped Eu
(base) kveeresh@KS-MacBook-Air-2 Computer-Science
• cd /Users/kveeresh/Desktop/Computer-Science ;
  /usr/bin/env /Users/kveeresh/Library/Java/JavaV
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6/redhat_java/jdt_ws/Computer-Science_4e282deb/bi
n-Loops.Patterns2
Printing Odd Pattern
* * * * *
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* * * * *
* * * * *
```

Skipped even place

$$N=5$$

The screenshot displays a Java IDE with a dark theme. The main editor window shows the following code:

```
30 public static void printStarsOnlyAtBorders(int n){
31
32     System.out.println(x:"Printing Stars At Border");
33
34     for(int i = 0 ; i < n ; i++){
35
36         for(int j = 0 ; j < n ; j++){
37
38             if(i==0 || j==n-1 || i==n-1 || j==0 ){
39
40                 System.out.print(s:"* ");
41             }
42             else{
43                 System.out.print(s:" ");
44             }
45         }
46         System.out.println();
47     }
48     System.out.println();
49 }
50
51
52
53 Run | Debug | Run main | Debug main
54 public static void main(String[] args) {
55
56     printStarOdd(n:7);
57
58     printStarsOnlyAtBorders(n:7);
59 }
```

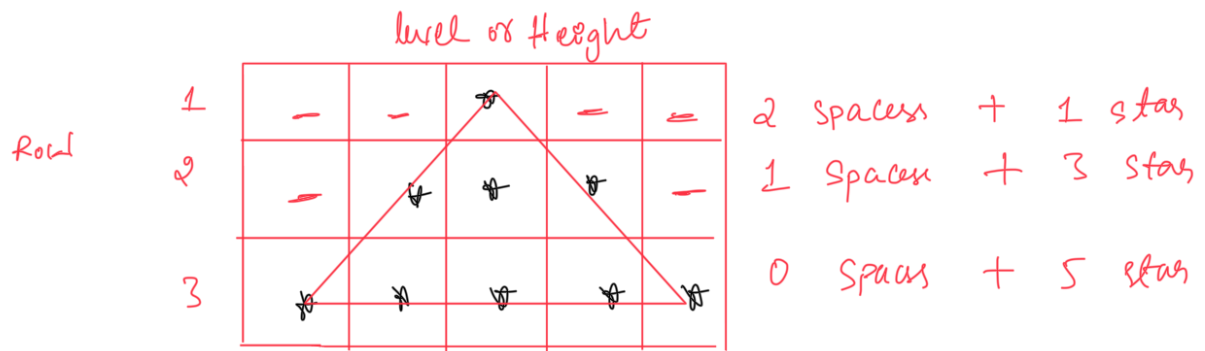
To the right of the code editor, there is a terminal window showing the output of the program:

```
(base) kveeresh@KS-MacBook: % cd /Users/kveeresh/De
% /usr/bin/env /Users/kvee
rtua/Machines/openjdk-24.
va -agentlib:jdwp:transpo
spenday,address=localhost
-XX:+ShowCodeDetailsInExc
s/kveeresh/Library/Applic
/workspaceStorage/2ecbf2b
6/nrdhat.java/jdt_ws/Comp
n Loops.Patterns2
Printing Odd Pattern
* * * * *
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* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
Printing Stars At Border
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

note $\theta_1 = 90^\circ$

$g = \text{columns}$

Pyramid Pattern



```

J Patterns.java  J Patterns2.java 2, U x  J Patterns.class
> J Patterns2.java > Language Support for Java(TM) by Red Hat > Patterns2 > printPyramid(in
4 public class Patterns2 {
54     public static void printPyramid(int n) {
55
56         System.out.println(x:"Printing Star Pattern");
57
58         int width = 2 * n - 1;
59
60         for (int row = 0; row < n; row++) {
61
62             for (int col = 0; col < width; col++) {
63
64                 if (col >= (n - row - 1) && col <= (n + row - 1)) {
65                     System.out.print(s:"*");
66                 } else {
67                     System.out.print(s:" ");
68                 }
69             }
70             System.out.println();
71         }
72     }
73
74     System.out.println();
75 }
76
77 Run | Debug | Run main | Debug main
78 public static void main(String[] args) {
79
80     printStarOdd(n:7);
81
82     printStartsOnlyAtBorders(n:7);
83
84     printPyramid(n:7);
85
86

```

```

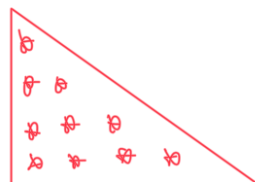
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

Printing Stars At Border
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

Printing Star Pattern
*
***
*****
*****
*****
*****
*****
*****

```

Pattern



```

76
77 public static void PrintStarTriagleLeft2 ( int size )
78 {
79     System.out.println(x:"print triangle at Left");
80
81     for ( int row =1; row <= size ; row ++ )
82     {
83         for ( int col =1; col <= row ; col ++ )
84         {
85             System.out.print( s:" * " ) ;
86         }
87         System . out . println ( ) ;
88     }
89     System.out.println();
90 }

```

```

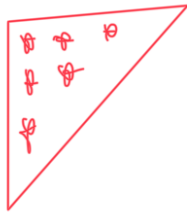
*
***
*****
*****
*****
*****
*****

print triangle at Left
*
* *
* * *
* * * *
* * * * *
* * * * *
* * * * *

(base) kveeresh@Ks-MacBook-Air-2

```

Pattern



```

91
92 public static void PrintStarTriangleExp2 ( int size )
93 {
94     System.out.println(x:"Print Traigle Inverted and left aligned");
95     for ( int row = size ; row >= 1; row-- )
96     {
97         for ( int col = row ; col >= 1; col-- )
98         {
99             System.out.print ( s:" * " );
100         }
101         System . out . println ( ) ;
102     }
103 }
104
105

```

print triangle at Left

```

*
* *
* * *
* * * *
* * * * *
* * * * * *

```

Print Traigle Inverted and left aligned

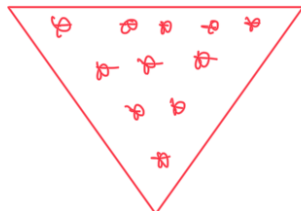
```

* * * * * * *
* * * * * *
* * * * *
* * * *
* * *
* *
*

```

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Pattern



Inverted pyramid pattern

```

105
106 public static void printInvertedPyramid(int rows) {
107
108     System.out.println(x:"Print Inverted Pyramid ");
109
110     for (int i = rows; i >= 1; i--) {
111
112         for (int space = 0; space < rows - i; space++) {
113             System.out.print(s:" ");
114         }
115
116         for (int star = 0; star < (2 * i - 1); star++) {
117             System.out.print(s:"* ");
118         }
119
120         System.out.println();
121     }
122 }
123
124
125

```

print triangle at Left

```

*
* *
* * *
* * * *
* * * * *
* * * * * *

```

Print Traigle Inverted and left aligned

```

* * * * * * *
* * * * * *
* * * * *
* * * *
* * *
* *
*

```

Print Inverted Pyramid

```

*****
*****
*****
*****
*****

```

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Java Notes: Star Pattern Printing

September 14, 2025

1 Introduction

In this note, we explore different **star printing patterns** in Java using nested loops. Each function demonstrates a specific approach to working with **for** loops to control rows and columns.

2 Java Code

```
1 public class StarPatternPrinting {
2
3     public static void main(String[] args) {
4
5     }
6
7     public static void PrintGrid1(int size)
8     {
9         for (int row=1; row <= size; row++)
10        {
11            for (int col=1; col <= size; col++)
12            {
13                System.out.print(row + "," + col + " ");
14            }
15            System.out.println();
16        }
17    }
18
19    public static void PrintStarGrid2(int size)
20    {
21        for (int row=1; row <= size; row++)
22        {
23            for (int col=1; col <= size; col++)
24            {
25                System.out.print("*");
26            }
27            System.out.println();
28        }
29    }
30 }
```

```

31 public static void PrintStarTriagleLeft2(int size)
32 {
33     for (int row=1; row <= size; row++)
34     {
35         for (int col=1; col <= row; col++)
36         {
37             System.out.print("*");
38         }
39         System.out.println();
40     }
41 }
42
43 public static void PrintStarTriagleExp2(int size)
44 {
45     for (int row=size; row >= 1; row--)
46     {
47         for (int col=row; col >= 1; col--)
48         {
49             System.out.print("*");
50         }
51         System.out.println();
52     }
53 }
54
55 }

```

3 Explanation of Methods

3.1 PrintGrid1

- Prints a coordinate grid of size $n \times n$.
- Each cell contains its row and column index.

Example for size = 3:

```

1,1 1,2 1,3
2,1 2,2 2,3
3,1 3,2 3,3

```

3.2 PrintStarGrid2

- Prints a solid square grid of stars.
- Size determines both width and height.

Example for size = 3:

```

***
***
***

```

3.3 PrintStarTriagleLeft2

- Prints a left-aligned right-angled triangle of stars.
- Row i has exactly i stars.

Example for size = 4:

```
*
**
***
****
```

3.4 PrintStarTriagleExp2

- Prints an inverted right-angled triangle of stars.
- Row i starts with i stars and decreases with each row.

Example for size = 4:

```
****
***
**
*
```

4 Conclusion

These patterns are a foundation for mastering **loops** and are widely asked in coding interviews. They help in developing logic for nested iterations, grid traversal, and problem-solving in **DSA**.